WellWatch Syria: Idea Proposal

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1. Project idea

Much of Syria's irrigated farming depends on groundwater, and it is being pumped faster than it refills. In Hasakah, a local geologist estimates groundwater is dropping 2 to 3 m a year (North Press, 2025). With less river water because of drought, upstream dams and damaged infrastructure, farmers rely more on wells, many now run by solar pumps. About 1.5 million people in Hasakah and its countryside have faced water shortages since 2019 (Co-operation in Mesopotamia, 2026), and some villages are emptying: in Tel Sakra only 18 of 55 families remain (Syriac Press, 2025).

Our question is: where is irrigated farming growing, or holding steady through dry years, in areas that likely depend on groundwater, and where should agencies check first? There is no up-to-date map that answers this today.

We will build a screening tool for Hasakah as a pilot. It maps irrigated, rainfed and fallow fields from satellite data, tracks how they change each year, adds drought data, and ranks zones for field visits. The users are the water and agriculture directorates and the NGOs and UN agencies that fund wells, pumps and drip irrigation; the farming families who depend on these wells benefit. The map shows irrigation pressure, not groundwater levels, since satellites cannot tell if water comes from a well or a canal. We mask land near rivers and canals and treat the results as places to check.

2. Relevance to the SDGs

Knowing where irrigation pressure is highest lets agencies place groundwater monitoring and drip irrigation support there first (SDG 6.4, water scarcity). Helping farms keep producing in dry years supports SDG 2.4 (sustainable food production) and SDG 13.1 (resilience to drought).

3. Literature examples

Mhanna, Halloran and Brunner (2024), Groundwater in armed conflict zones: the case of the Orontes River Basin, Syria. With almost no field data, they used machine learning on satellite images to measure irrigated land lost after 2011 and radar (InSAR) to link it to groundwater recovery. This shows the analysis works in Syria; we apply it to the northeast.

Bolognesi et al. (2020), Harmonized Landsat 8 and Sentinel-2 time series data to detect irrigated areas, Remote Sensing 12(8), 1275. They mapped irrigated land in southern Italy from NDVI time series and compared six classifiers; random forests, SVMs and boosted trees did best (about 90% accuracy). This is our starting method.

4. Candidate data

Sentinel-2 (2017 to 2026) for NDVI and NDMI, Sentinel-1 radar, Google AlphaEarth satellite embeddings (2017 to 2025), CHIRPS rainfall and ERA5 temperature, JRC Global Surface Water and OpenStreetMap for rivers and canals, and the ESA World cereal 2021 irrigation map for comparison. We will label about 400 fields by hand in Google Earth Pro.

5. Proposed approach

We start with a small area near Qamishli and al-Malikiyah. For each field we build monthly NDVI, NDMI and radar values and train a Random Forest to classify it as irrigated, rainfed or fallow, compared with a simple NDVI threshold and a model on AlphaEarth embeddings. We test on areas and years the model has not seen. Then we map irrigated area per year, combine it with rainfall and temperature, and rank zones where irrigation stayed high in dry years. The output is an interactive map showing why each zone was flagged and how confident the model is.